Royal Devon University Healthcare NHS Foundation Trust

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Clinical Guideline for:
New incidental pulmonary hypertension (PHT) found on pre-operative echocardiogram

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SUMMARY

This guideline outlines a step by step approach to guide anaesthetic clinicians on assessing the severity of incidental pulmonary hypertension documented on pre-operative echocardiogram, including risks, management and suitability for care in a non-pulmonary hypertension institution.

KEY POINTS

The essential elements of this guideline are:

1. Assessment of right ventricular physiological reserve
2. Preoperative optimisation, contingent on understanding the cause
3. Perioperative monitoring and support of RV and pulmonary pressures
4. Appreciation of need to support RV postoperatively as well

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1. INTRODUCTION

Echocardiography, ordered for other reasons often comments on the likelihood of pulmonary hypertension. This is often an incidental finding in patients with no prior knowledge or symptoms and on no treatment for this condition. With basic structured assessment, the majority of these patients can be safely managed perioperatively with some core basic principles, with little risk of right ventricular collapse. The below guideline should assist in the management in this very common group of patients.

2. BACKGROUND

2.1 This is a new clinical guideline.

3. DEFINITIONS

- CTEPH: Chronic Thrombo-Embolic Pulmonary Hypertension
- ECMO: Extra-Corporeal Membrane Oxygenation
- EF: Ejection Fraction
- FAC: Fractional Area Change (surrogate for EF)
- HFpEF: Heart Failure with Preserved Ejection Fraction (diastolic)
- HFrEF: Heart Failure with Reduced Ejection Fraction (systolic)
- IVC: Inferior Vena Cava
- IVS: Inter Ventricular Septum
- LA: Left Atrium
- LV: Left Ventricle
- MPAP: Mean Pulmonary Artery Pressure
- PAH: Pulmonary Arterial Hypertension
- PCWP: Pulmonary Capillary Wedge Pressure
- PAH: Pulmonary Arterial Hypertension
- PH: Pulmonary Hypertension
- PVD: Pulmonary Vasodilators
- PVR: Pulmonary Vascular Resistance
- RA: Right Atrium
- RV: Right Ventricle
- SPAP: Systolic Pulmonary Artery Pressure
- TAPSE: Tricuspid Annular Plane Systolic Excursion
- TR: Tricuspid Regurgitation
- WU: Woods Units

4. MAIN BODY OF GUIDELINE

4.1 Pulmonary systolic pressures rise by 1mmHg per year over the age of 65[i]. Diagnosis of pulmonary hypertension is now defined as mean pulmonary artery pressures over 20mmHg[ii] thus it is increasingly likely to be diagnosed on a preoperative echocardiogram done for other reasons.
4.2 Surgical patients over the age of 65 with PHT will be highly unlikely to be offered extracorporeal membrane oxygenation (ECMO) or any other cardiac mechanical support and thus will normally be managed in a non-cardiac centre.

The key to safe perioperative management is the assessment of right ventricular (RV) reserve, from history, examination and investigations. A RV with normal size and function is extremely unlikely to decompensate acutely if basic supportive measures are applied.

4.3 True assessment of PHT requires right heart catheterisation to determine whether it is caused by pulmonary pathology (pre-capillary), left heart disease (post-capillary) or a combination of both. This is of vital importance as the management strategies will be different for each type. Other investigations will provide additional clues as to the origin and cause of the PHT.

4.4 Incidental PHT is discovered on an echocardiogram predominantly by scrutinising tricuspid regurgitation (TR). The TR jet velocity can be measured in metres/second and pulmonary pressures inferred by the Simplified Bernoulli Equation, whereby systolic pulmonary artery pressure SPAP = $4x\ (TR)^2 + CVP$. The principle is that the RV ejects blood forwards into the main PA and backwards (as TR) with the same velocity and pressure. The retrograde TR jet will be blunted by the right atrial pressure (CVP) thus this has to be added on.

Thus, PA pressures will be higher than the number calculated as CVP has to be added. Also, if the patient has no TR then the confidence in determining the presence of PHT is much less.

The schematic below is an attempt to predict the safety of proceeding with surgery in our non-cardiac centre.

Most likely, there will be a mixture of positive and negative signs, thus a balance must be made regarding safety. However, the demonstration of right ventricular reserve plus a cardio-stable anaesthetic should make the potential for RV sudden decompensation very unlikely

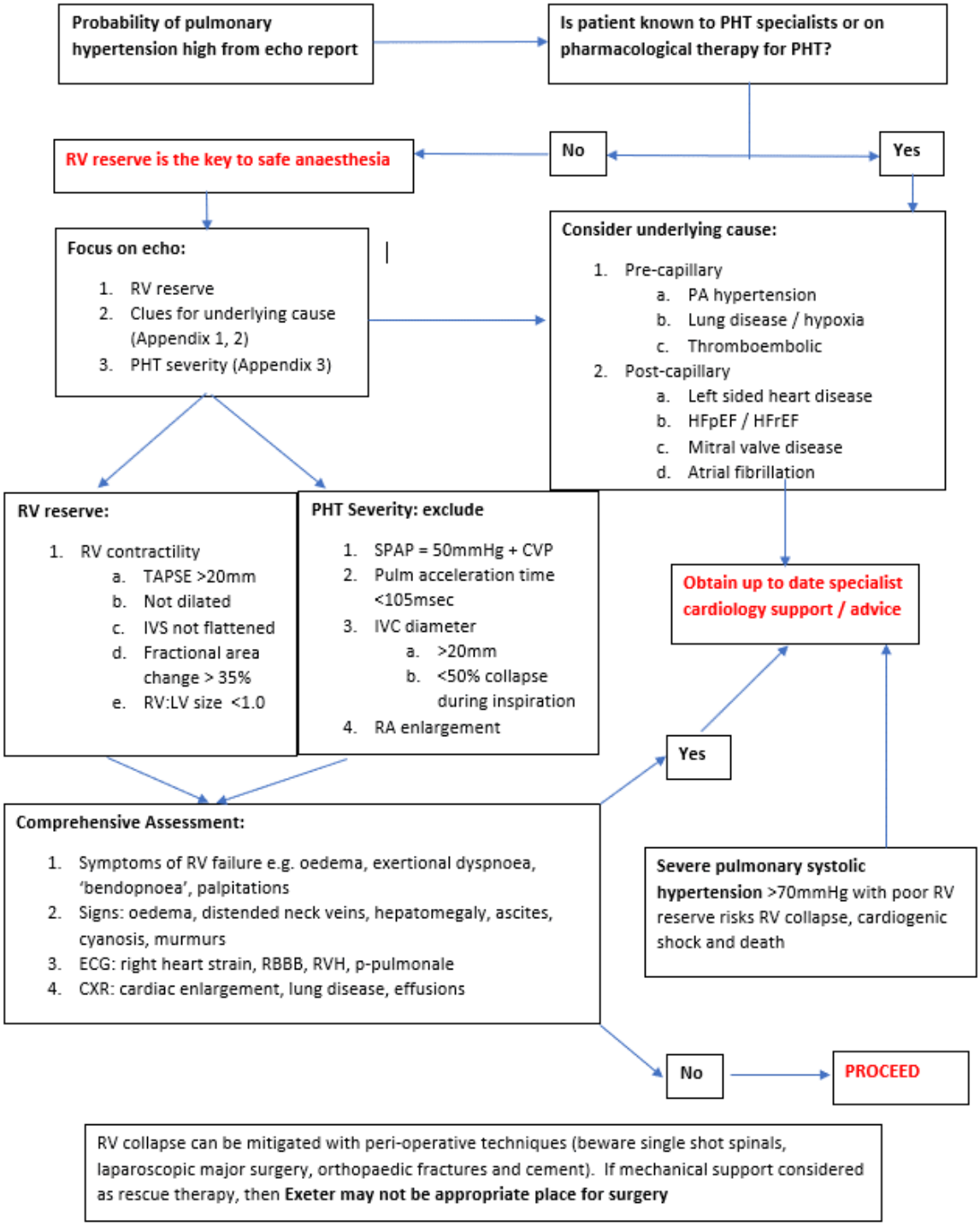
Also attached is a summary of perioperative considerations and guidance taken from a bigger prep for surgery document on pulmonary hypertension

5. MONITORING COMPLIANCE WITH THIS GUIDELINE

5.1 Compliance with this guideline will be managed by anaesthetics and theatres

6. ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

Lam CSP, Borlaug BA, Kane GC, et al. Age-associated increases in pulmonary artery systolic pressure in the general population. Circulation. 2009;119(20):2663–2670.
[1] Evaluation and Management of Pulmonary Hypertension in Noncardiac Surgery: A Scientific Statement From the American Heart Association. Circulation Volume 147 • Number 17 • 25 April 2023 Pages: 1317 – 1343 PubMed: 36924225


Probability of pulmonary hypertension high from echo report
Is patient known to PHT specialists or on pharmacological therapy for PHT?
No
Yes
RV reserve is the key to safe anaesthesia
Focus on echo:
1. RV reserve
2. Clues for underlying cause (Appendix 1, 2)
3. PHT severity (Appendix 3)
Consider underlying cause:
1. Pre-capillary
a. PA hypertension
b. Lung disease / hypoxia
c. Thromboembolic
2. Post-capillary
a. Left sided heart disease
b. HFpEF / HFrEF
c. Mitral valve disease
d. Atrial fibrillation
RV reserve:
1. RV contractility
a. TAPSE >20mm
b. Not dilated
c. IVS not flattened
d. Fractional area change > 35%
e. RV:LV size <1.0
PHT Severity: exclude
1. SPAP = 50mmHg + CVP
2. Pulm acceleration time <105msec
3. IVC diameter
a. >20mm
b. <50% collapse during inspiration
4. RA enlargement
Obtain up to date specialist cardiology support / advice
Yes
Comprehensive Assessment:
1. Symptoms of RV failure e.g. oedema, exertional dyspnoea, 'bendopnoea', palpitations
2. Signs: oedema, distended neck veins, hepatomegaly, ascites, cyanosis, murmurs
3. ECG: right heart strain, RBBB, RVH, p-pulmonale
4. CXR: cardiac enlargement, lung disease, effusions
Severe pulmonary systolic hypertension >70mmHg with poor RV reserve risks RV collapse, cardiogenic shock and death
No
PROCEED
RV collapse can be mitigated with peri-operative techniques (beware single shot spinals, laparoscopic major surgery, orthopaedic fractures and cement). If mechanical support considered as rescue therapy, then Exeter may not be appropriate place for surgery

Summary Guideline for Intra-operative Management

Pre-operative history, examination and investigations to establish RV reserve and severity of PHT

Is Exeter appropriate place to operate?

Get specialist assistance if RV reserve compromised

Things To Avoid

1. sedative premedication
2. single shot spinals (consider epidural)
3. unopposed alpha agonists e.g. phenylephrine

Vigilance

1. Laparoscopic surgery
2. Fractures, cement
3. Fluid shifts, blood loss
4. Head up / down position

Ventilation strategy

1. Minimise hypoxia, hypercarbia, acidosis, extreme lung volumes
2. Balance Supra-glottic PSV ventilation against ETT + IPPV

Haemodynamic support:

1. Milrinone + low dose noradrenaline / vasopressin
2. Dobutamine + low dose noradrenaline / vasopressin
3. Adenaline
4. Avoid unopposed alpha-agonists

Monitoring:

1. Arterial line – always
   a. Frequent gases
2. Central line – moderate, major ops
   a. CVP – number and trace
   b. Central O_2 sats
3. PA catheter – major ops, if expertise available
4. TOE / TTE – major ops if expertise available

Induction Agents

1. Cardio-stable induction
2. Propofol, ketamine caution
3. Consider etomidate
4. Fentanyl
5. Muscle relaxant – avoid histamine release (?atracurium), but allergy risk of rocuronium – vecuronium or cis-atracurium safest, unless RSI

Stable: PVR, MAP, Sinus Rhythm, normovolaemia

Consider Intensive Care Support for ALL but minor cases

RV may remain threatened for a considerable time post operatively, e.g 72 hours

7. APPENDICES

Appendix 1: aetiology

Group 1 Pulmonary arterial hypertension (PAH). Idiopathic PAH, connective tissue disease associated PAH, and congenital heart disease (mainly Eisenmenger syndrome) are the most common subgroups; others include porto-pulmonary, HIV, and drug/toxin-induced PAH.

Group 2 PH associated with increased pulmonary capillary wedge pressure (post-capillary PH), due to left-sided heart disease.

Group 3 Lung disease and chronic hypoxia-related PH.

Group 4 Predominantly chronic thromboembolic PH (CTEPH).

Group 5 PH caused by multisystemic disorders or *multiple/unknown mechanisms*, including sarcoidosis and haematological conditions.

Appendix 2: treatment by cause

Group 1: pulmonary vasodilators
Group 2: treat left sided heart disease
Group 3: treat lung disease
Group 4: anticoagulation, pulmonary endarterectomy
Group 5: underlying cause if known

Appendix 3: PHT severity grading

Definition	Haemodynamic characteristics
PH	MPAP > 20mmHg
Pre-capillary PH	MPAP > 20mmHg PAWP ≤ 15mmHg PVR > 2WU
Isolated post-capillary PH	MPAP > 20mmHg PAWP > 15mmHg PVR ≤ 2 WU
Combined post- and pre-capillary PH	MPAP > 20mmHg PAWP > 15mmHg PVR > 2 WU

PVR measured in dynes-sec/cm[5]
= 80 x (Mean Pulmonary Arterial Pressure in mmHg – Left Atrial Pressure in mmHg) . /. Pulmonary Flow in L/min

Normal PVR is between 20 and 130 dynes-sec/cm[5]

PVR measured in mmHg-min/L WOOD UNITS
= (Mean Pulmonary Arterial Pressure in mmHg – Left Atrial Pressure in mmHg) ./. Pulmonary Flow in L/min

Normal WU is < 3